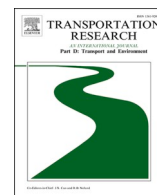




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Environment-economic comparison of potential alternative fuel heavy-duty trucks in China

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ABSTRACT

Heavy-duty trucks, as important players in pollution and greenhouse gas emissions, need to undertake critical responsibility in road freight. By estimating the well-to-wheel emissions and total cost of ownership for internal combustion engine heavy-duty trucks, this study proposes the model of carbon-abatement benefit to conducting an environment-economic comparison of traditional diesel and potential alternative fuel. The results show that liquefied natural gas has better economic performance, with a cost 14.9% lower than diesel, and biodiesel reduces greenhouse gas emissions by more than 40% compared to diesel. Compared to other alternative fuels, liquefied natural gas has better carbon reduction benefits, with a carbon abatement value of 0.51 CNY/km. By 2040, all alternative fuels have positive carbon abatement values, ranging from 0.26 to 1.28 CNY/km. The progress of replacing diesel by alternative fuels for heavy duty trucks depending on the incentive policies, the clean production of fuels, and the competition from electric trucks.

1. Introduction

1.1. Background

Road freight plays a vital role in global cargo transportation, accounting for 65% of cargo transportation emissions in 2021 (ITF, 2021), which undertakes the vast majority of China's freight transportation demand, with a contribution of 74% in 2021 (National bureau of statistics of China, 2022). In China, as shown in Fig. 1, heavy-duty trucks (HDTs) contributed the majority of the NOx and PM emissions from all vehicles in 2019, accounting for 74% and 52% respectively (P.R.C., 2020).

Electric vehicles (EVs) are regarded as the main pathway to reduce GHG emissions in the transport sector, but their development is still limited by some deficiencies, such as battery thermal management (Zhao et al., 2019) and low-temperature performance (Wang

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Nomenclature

B5	The proportion of biodiesel mixed in fossil diesel is 5%
B20	The proportion of biodiesel mixed in fossil diesel is 20%
B100	Pure biodiesel
BEVs	Battery Electric Vehicles
CAB	Carbon-abatement Benefit
CB H ₂	Coal-based Hydrogen
EVs	Electric Vehicles
GHG	Greenhouse Gas
HDTs	Heavy-Duty Trucks
H/NP H ₂	Hydroelectric/Nuclear Power to Produce Hydrogen
H2ICE	Hydrogen Internal Combustion Engine
IBP H ₂	Industrial By-product Hydrogen
NGB H ₂	Natural Gas-based Hydrogen
PTW	Pump-To-Wheel
WTP	Well-To-Pump
WTW	Well-To-Wheel
TCO	Total Cost of Ownership

et al., 2018). The operation performance of EVs in northern China (north of 35°) is poor as their daily extremely low-temperature threshold range in winter from −16 °C~−40 °C (Wang et al., 2023). However, the performance of traditional internal combustion engine vehicles is less affected in low-temperature environments. In cold regions, using alternative fuels for internal combustion engine HDTs can reduce GHG emissions in the freight sector and maintain better vehicle operating efficiency.

However, factors such as fuel reliability (Ghadikolaei et al., 2021), mileage anxiety (Gnann et al., 2022), and operating costs (Cantillo et al., 2022) will hinder the development of alternative fuel HDTs. Based on the background of energy conservation and emission reduction, government policies and regulations, as well as the environmental awareness of enterprises, provide powerful driving forces for the adoption of alternative fuel HDTs. Alternative fuels could improve engine efficiency (Bae and Kim, 2017) and have the potential to resolve pollution (Abdul Karim and Khan, 2018).

As the main carrier supporting road freight, HDTs have become the key object of emission control in the transport sector. Simultaneously, the adoption of EVs in cold regions is limited by battery technology and low efficiency under extreme operating conditions (Zhou et al., 2023), while the performance of internal combustion engine vehicles is relatively stable. Therefore, China's low-carbon development must find a suitable path for alternative fuel HDTs, especially in the cold regions of northern China during winter. The study aims to analyze the economic and environmental performances of four promising HDT alternative fuel pathways, including liquefied natural gas (LNG), biodiesel, hydrogen, and methanol, selected based on the characteristics of China's industrial development. Four hydrogen production methods will be considered, including natural gas-based hydrogen (NGB H₂), coal-based hydrogen (CB H₂), industrial by-product hydrogen (IBP H₂), and hydroelectric/nuclear power to produce hydrogen (H/NP H₂).

1.2. Literature review

The literature review found that the key to considering the use of alternative fuel HDT instead of traditional diesel HDT lies in the

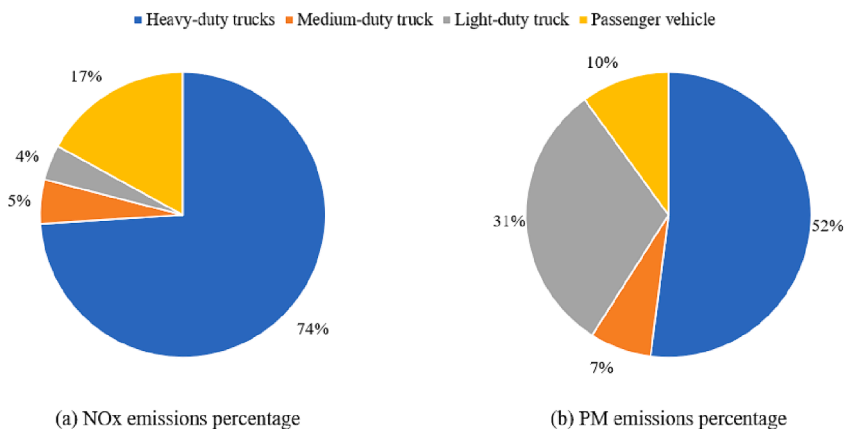


Fig. 1. 2019 NOx and PM Emissions percentage of different vehicle types in China.

cost and benefits of fleet operators, namely total cost of ownership (TCO) analysis and comparison of wheel-to-wheel emissions (WTW). WTW covers the pathway from resource extraction to use in vehicles but excludes the life cycle of the vehicle itself (Prussi et al., 2020). On the one hand, compared to the passenger car industry, HDTs focus more on the total cost of vehicles throughout their entire service life (Noll et al., 2022). On the other hand, adopting alternative fuels needs to consider whether the entire fuel cycle is clean (Singh Verma et al., 2022).

Some scholars consider the economics of different alternative fuels HDTs based on TCO, which considers all direct and indirect costs throughout the entire lifecycle. Noll et al. studied the relative cost competitiveness of HDTs with different alternative driving technologies in Europe based on TCO, and found that LNG vehicles have the best economic benefits among different alternatives, and the cost is lower than EVs (Noll et al., 2022). Hydrogen internal combustion engine (H₂ICE) vehicles have the same prospects as hydrogen fuel cell vehicles in reducing GHG emissions (Gillingham, 2007), with the expensive cost of hydrogen production and transportation (Johnston et al., 2022; Sens et al., 2022). Compared with gasoline vehicles, methanol vehicles have promising economic and environmental benefits (Luo et al., 2022). With the improvement of methanol production technology, the operation and maintenance cost of methanol vehicles can continue to decrease.

Other scholars consider the entire lifecycle of fuel from production to use and explore the WTW emissions of different alternative fuels HDTs. Compared with diesel fuel, HDTs using LNG fuel can reduce GHG emissions by about 8% (Song et al., 2017), which has the advantage of pollutant emissions reduction (Hao et al., 2021; Zheng et al., 2022); HDTs using hydrogen can effectively reduce the emissions of CO, CO₂ and exhaust smoke (Shadidi et al., 2021); HDTs using methanol as fuel can reduce NO_x, smoke and PM emissions of vehicles (Tian et al., 2022). Martínez et al. conducted emission experiments on biodiesel vehicles using waste edible oil as feedstock and found that compared to pure diesel, biodiesel fuel has a smaller GHG emission factor (Martínez-Martínez et al., 2022). Biodiesel has become a suitable vehicle fuel in the aspect of the environment (Karavalakis et al., 2017; Wong et al., 2022) and resource reuse (Chung et al., 2019). Prussi et al. evaluated the WTW energy use and GHG emissions for various potential powertrains and fuel options in Europe and found that the conversion route and feedstock chosen have a significant impact on the final energy consumption and GHG emissions (Prussi et al., 2020).

Based on the analysis of TCO and WTW emissions, some literature proposes factors that need to be considered or further expanded in the development of alternative fuel trucks. Gustafsson et al. evaluated the WTW emissions of different alternative fuels used in heavy-duty vehicles, such as compressed natural gas, electricity, hydrogen, biomethane, and hydrotreated vegetable oil, and found that energy carriers with high electricity dependence may not necessarily be better than diesel from the perspective of WTW (Gustafsson et al., 2021). Alonso-Villar et al. expanded energy safety to evaluate different alternative fuel trucks, including biodiesel, LNG, renewable diesel, and hydrogen fuel cells, and believe that developing incentive measures to reduce fuel costs at gas stations is very effective in promoting fuel cell vehicles and LNG (Alonso-Villar et al., 2022). Bae et al. explored the decision of HDT fleet operators to adopt alternative fuels and found that the reasons for not adopting the decision include inappropriate functionality, reliability/safety issues, unacceptable financial costs, or increased operational complexity due to insufficient refueling infrastructure (Bae et al., 2022).

Unfortunately, few studies have quantitatively compared the TCO and WTW emissions of four types of alternative fuel HDTs selected in this study (LNG, biodiesel, hydrogen, and methanol). Moreover, the influence of vehicle related costs, carbon tax subsidies, electricity production systems, and the feedstock used for alternative fuels have not been fully considered. Due to the differences in industrial development status, fuel prices, hydrogen fuel production modes, and efficiency, the alternative fuel pathways of HDTs may be different in various regions (Mingolla and Lu, 2021). This study will conduct an environment-economic comparison of potential alternative fuel heavy-duty trucks in China.

2. Methodology

To better analyze the emission abatement pathways of HDT, this study establishes an environment-economic analysis framework, comparing the economic benefits (TCO) and environmental impacts (WTW emissions) of traditional diesel and different alternative fuel HDTs, including LNG, biodiesel, hydrogen, and methanol, attempting to provide a reference for policymakers and relevant professionals in the transport sector.

2.1. Total cost of ownership

TCO can reflect the most economical fuel to replace traditional diesel for HDTs and includes four major cost components: infrastructure construction, vehicle purchase, operation, and other related (i.e., road rent, parking, repair & maintenance, and annual inspection). It is worth noting that the TCO in the study is a generalized model that specifically considers infrastructure costs. TCO of potential alternative fuel HDTs in the study is calculated according to Eq. (1) (Zhang et al., 2022):

$$TCO = \frac{C_i}{n} + C_b + \sum_{t=1}^8 \frac{R_e \times P_f \times L + C_m}{(1+r)^t} \quad (1)$$

In which, C_i refers to the infrastructure construction cost, C_b is the trucks' purchase price, n is the truck service capacity of the infrastructure. R_e refers to the fuel consumption per kilometer, P_f is the purchase price of different fuels, L is the trucks' annual driving mileage. C_m refers to other related costs. r refers to the discount rate and sets it at 10%, t is the time step and assumes the study period of 8 years for the truck's lifetime.

2.2. Well-to-wheel emissions

The life cycle of a truck is divided into a vehicle cycle and a fuel cycle. The vehicle cycle is the process of manufacturing, assembling, using, and scrapping trucks. The fuel cycle (WTW) is divided into well-to-pump cycle (WTP) and pump-to-wheel cycle (PTW), as shown in Fig. 2, including the fuel feedstock extraction, the fuel production and refueling, and the vehicle operation (Ren et al., 2020). All the HDT technologies considered in this study are based on internal combustion engine, except for tanks for liquefied natural gas and hydrogen, which are similar. Due to the limited research on vehicle cycle emissions, this study ignored the differences in vehicle cycle emissions and only focused on the pollution emissions of fuel WTW, including GHG, NOx, and PM2.5. In the study, GHG emissions include CO₂, methane, and nitrous oxide, which are the most important GHGs. All HDTs using potential alternative fuels and diesel are assumed to operate under the same conditions.

The calculation formula of the vehicle WTW total emission E_{wij} is shown in Eq. (2) (Ma et al., 2023).

$$E_{wij} = E_{pij} + E_{uij} \quad (2)$$

E_{wij} refers to the total amount of the j -type WTW emissions of the i -type alternative fuel technologies. Where, the emissions include GHG, NOx, and PM2.5, and alternative fuel HDTs include LNG trucks, biodiesel trucks, H₂ICE trucks, and methanol trucks.

The calculation formula of WTP stage emission is shown in Eq. (3).

$$E_{pij} = M \times E_{ci} \times F_{pij} \quad (3)$$

E_{pij} refers to the j -type WTP emissions of the i -type alternative fuel technologies. M refers to the truck's life-cycle mileage, E_{ci} refers to the fuel consumption per kilometer of trucks, F_{pij} is the emission factor for the WTP cycle.

The calculation formula of PTW stage emission is shown in Eq. (4).

$$E_{uij} = M \times F_{ij} \quad (4)$$

E_{uij} refers to the j -type PTW emissions of the i -type alternative fuel technologies. F_{ij} is the emission factor of the PTW stage.

2.3. Carbon-abatement benefit

GHG emissions can be reduced in various ways, including improving vehicle operation efficiency, using cleaner fuels, and improving post-treatment systems. Any form of GHG emission reduction measures will bring corresponding costs, which are related to technological changes (Bralower and Bice, n.d.; Du et al., 2022). China opened the carbon trading market in 2021, utilizing market mechanisms to control and reduce GHG emissions (State Council P.R.C., 2021).

To better compare the alternative fuel HDTs, this study proposes a calculation method of carbon-abatement benefit (CAB) that combines the carbon price and vehicle replacement cost, as shown in Eq. (5). CAB refers to the benefits that can be obtained per kilometer of vehicle operation under the background of replacing traditional diesel trucks with alternative fuel trucks and calculating GHG emission reduction.

$$CAB = \frac{P_C(E_{CD} - E_{Ci}) - (TCO_i - TCO_D)}{M} \quad (5)$$

Where, P_C is the carbon price, based on the data of the Beijing carbon trading platform, the average carbon trading cost (74.60 CNY/ton) on January 12, 2022, was selected as the carbon tax value in the study (Platform, 2022). E_{CD} is the GHG emission value of traditional diesel HDTs, E_{Ci} refers to the GHG emission value of alternative fuel HDTs. TCO_i is the TCO value of the alternative fuel HDTs, TCO_D refers to the TCO value of traditional diesel HDTs. M refers to the trucks' life-cycle mileage.

3. Data

The most representative relevant data was collected and screened in the study, including TCO parameters and WTW emissions data for different alternative fuel HDTs, which are derived from second-hand data based on literature research and field investigations. Due to limited public data on alternative fuel HDTs in China, some data is sourced from other countries.

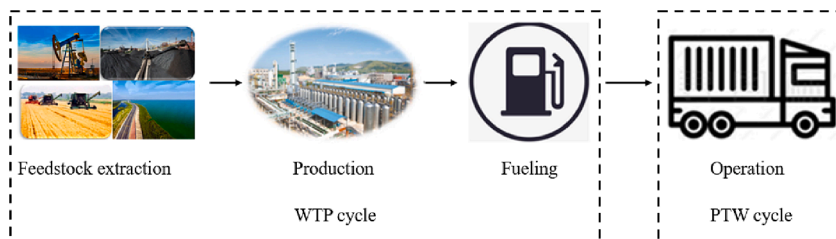


Fig. 2. WTW flow chart of vehicles' fuel.

3.1. TCO parameters for different alternative fuel HDTs

The cost parameters of different alternative fuel HDTs are shown in Table 1. Among them, the cost of infrastructure is mainly the fuel filling station, and the data are all from AFLEET background data, which has good comparability. Based on the selected price (6.82 CNY/L) of Beijing NO.0 diesel on December 10, 2021. According to the relevant models in the current market, this study can get the purchase costs of different alternative fuel HDTs. Other related expenses such as insurance, repair, and maintenance of HDTs are borne based on the actual operational status of China's transport sector.

3.2. WTW emissions data for different alternative fuel HDTs

To more effectively compare the environmental impact of different alternative fuel HDTs, this study analyzes their WTW emissions. According to the relevant opinions on carbon abatement and pollutant reduction regulations in China, GHG, NO_x, and PM_{2.5} are selected for the environmental impact analysis of alternative fuel HDTs in this study. WTW emissions data for different alternative fuel HDTs can be divided into two stages as follows.

Table 1

TCO parameters for potential alternative fuel HDTs.

Main	Component	Value	Unit	Source
Infrastructure (C_i)	Diesel	1,100,229	CNY	(Argonne National Laboratory, 2020)
	LNG	1,211,700	CNY	
	Biodiesel ^a	1,100,229	CNY	
	Hydrogen	8,607,600	CNY	
	Methanol ^b	1,409,100	CNY	
Service capacity (n) ^c	Fuel filling station	200	Number of vehicles	–
Energy consumption (R_e)	Diesel truck	0.36	Liter/km	(Yuan et al., 2018)
	LNG truck	0.66	Liter/km	
	B5 truck	0.36	Liter/km	(Shen et al., 2018)
	B20 truck	0.35	Liter/km	
	B100 truck	0.3	Liter/km	(Acar and Dincer, 2020; Department of Energy, U.S., 2020)
	Hydrogen truck	1.24	Liter/km	
Price of fuels (P_f)	Methanol truck	0.85 ^d	Liter/km	–
	Diesel	6.02 ^e	CNY/Liter	–
	LNG	3.71 ^f	CNY/Liter	–
	B5	5.95	CNY/Liter	(Shanghai Municipality, 2020)
	B20	6.48	CNY/Liter	(Department of Energy, U.S., 2021)
	B100	9.02	CNY/Liter	(Department of Energy, U.S., 2020; Nanhai District Government of Foshan City, 2021; Zhao et al., 2021)
	NGB H ₂	1.02	CNY/Liter	
	CB H ₂	0.73	CNY/Liter	
	IBP H ₂	1.23	CNY/Liter	
	H/NP H ₂	2.51	CNY/Liter	
	Methanol	2.13 ^g	CNY/Liter	–
Purchase price (C_b)	Diesel truck	436,500 ^h	CNY	–
	LNG truck	518,000 ⁱ	CNY	–
	Biodiesel truck	436,500 ^h	CNY	–
	Hydrogen truck	896,500 ^j	CNY	(Sun et al., 2014)
	Methanol truck	450,000 ^k	CNY	–
Other related costs (C_m)	Road rent	1.65	CNY/km	–
	Parking	2,000	CNY/year	–
	Repair & maintenance	10,000	CNY/year	–
	Annual inspection	1,000	CNY/year	–
Lifetime's driving mileage (M)	Vehicle's lifetime (t)	8	Years	(Hao et al., 2011; Ministry of Commerce P.R.C., 2013)
	Vehicle's driving mileage each year (L)	100,000	km/year	

^a The biodiesel infrastructure is the filling station, which is similar to diesel stations.

^b Based on the AFLEET Background data, they have the same equipment in the station, and the methanol infrastructure cost is assumed to be consistent with that of ethanol.

^c The service capacity of infrastructure is assumed based on the characteristics of different stations, which refers to the number of trucks that the infrastructure can serve per day.

^d Geely methanol HDTs consume about 85 L per 100 km.

^e Based on the petrochemical industry data in 2021, the average price of No. 0 diesel in Beijing is 6.02 CNY/L.

^f The average price of the LNG market in China in 2021 will be 5303.53 CNY/ton.

^g Based on the data of the China National Bureau of Statistics, the density of methanol is 0.792 g/cm³ and the price is 2694.9 CNY/ton.

^h The study assumes that biodiesel trucks cost the same as traditional diesel trucks because they have the same vehicle parts.

ⁱ The price of U460 LNG HDTs is 518,000 CNY.

^j The price of H₂ICE trucks is derived from the system transformation of diesel trucks.

^k The price of Geely methanol HDTs is 450,000 CNY.

(1) Emissions in the stage of fuel production (WTP)

The fuel production stage starts from obtaining energy feedstock to vehicle operation. The emissions of different alternative fuels in the WTP stage are shown in Table 2, including traditional diesel, LNG, biodiesel (B5, B20, B100), hydrogen fuels (NGB H₂, CB H₂, IBP H₂, H/NP H₂), and methanol. Different hydrogen production pathways bring different environmental impacts. For example, CB H₂ is a high-purity hydrogen gas prepared by a coal gasification and extraction process (Wang et al., 2020). Among the four hydrogen production pathways, HDTs with CB H₂ have the highest GHG and PM_{2.5} emissions due to the large emissions caused by coal treatment, especially high smoke and dust emissions. China has rich coal resources and mature related technologies, mainly using coal as the basic feedstock to produce methanol.

(2) Emissions in the stage of vehicle operation (PTW)

The operating phase of a vehicle refers to the process in which fuel is consumed during vehicle driving. The PTW stage emissions of different alternative fuel HDTs are shown in Table 2, including diesel trucks, LNG trucks, biodiesel trucks, and methanol trucks. Traditional diesel trucks have the highest unit GHG emissions in the PTW stage. Based on the combustion characteristics of hydrogen fuel, H₂ICE trucks have no GHG and PM_{2.5} emissions during the PTW phase, and their NO_x emissions are estimated by the test vehicle due to the limitation of available data.

4. Results

4.1. TCO comparison of different alternative fuel HDTs

According to the TCO parameters in Section 3.1, the TCO values of China's traditional diesel and alternative fuel HDTs can be calculated using Eq. (1), as shown in Fig. 3. The ranking of TCO results of different HDTs' pathways from low to high is as follows: LNG, methanol, CB H₂, B5, diesel, NGB H₂, B20, IBP H₂, B100, and H/NP H₂. Compared to traditional diesel trucks, the purchase price of LNG trucks increases by 18.7%, but because of lower vehicle operation costs, the TCO result of LNG HDTs is lower than that of diesel. The TCO for methanol trucks is lower than diesel due to the low cost of vehicle purchase and operation. Hydrogen fuel from different preparation routes has different fuel prices, and based on China's rich coal resources and mature coal industry, the hydrogen fuel cost of CB H₂ is the lowest among H₂ICE trucks.

The specific costs of HDT for different infrastructures are shown in Fig. 4. For infrastructure construction, China has the largest number of natural gas vehicles in the world, and the cost of LNG filling station is superior under the premise of large-scale application. Biodiesel and methanol filling stations can be directly transformed based on traditional diesel stations, resulting in lower infrastructure costs. However, the storage requirements of hydrogen fuel are rigorous, and the difficulty and cost of high-pressure storage and liquid storage are higher than other alternative fuels, so H₂ICE trucks have the most expensive infrastructure construction cost. It is worth noting that the infrastructure cost of trucks on different routes in the study is relatively small compared to other costs, as shown in Fig. 3.

4.2. WTW emissions of different alternative fuel HDTs

According to the emissions data in Section 3.2, the emissions of China's traditional diesel and alternative fuel HDTs, including GHG, NO_x, and PM_{2.5}, are calculated by using Eqs. (2), (3), and (4).

Table 2
Emissions factors of different alternative fuels HDTs in the stage of WTP and PTW.

Stage	Type	GHG	NO _x	PM _{2.5}	Units	Source
WTP	Diesel	19.44	0.0530	0.0033	g/MJ	(Ou et al., 2013; Society of Automotive Engineers, C., 2019)
	LNG	10.49	0.2685	0.0138	g/MJ	(Ou et al., 2013; Sharafian et al., 2019)
	B5	19.53	0.0881	0.0029	g/MJ	(Ou et al., 2013; Argonne National Laboratory, 2020)
	B20	19.81	0.0432	0.0026	g/MJ	
	B100	21.31	0.0343	0.0017	g/MJ	
	NGB H ₂	208.44	0.045	0.0058	g/MJ	(Lee et al., 2018; Ren et al., 2020; Wang et al., 2020)
	CB H ₂	318.47	0.03015	0.0315	g/MJ	
	IBP H ₂	93.36	0.03618	0.0189	g/MJ	
	H/NP H ₂	86.12	0.0076	0.0025	g/MJ	
	Methanol	188.13	0.3600	0.0119	g/MJ	(Xiang, 2017; Yao et al., 2018)
PTW	Diesel truck	952	12.1000	0.4100	g/km	(Tan et al., 2019; Yuan et al., 2018)
	LNG truck	709	7.7500	0.0199	g/km	(Wang et al., 2022)
	B5 truck	450	11.6600	0.4800	g/km	(Shen et al., 2018; Tan et al., 2019)
	B20 truck	435	10.5600	0.2000	g/km	
	B100 truck	480	9.4700	0.0900	g/km	
	H ₂ ICE truck	—	0.2217	—	g/km	(Zemo Partnership, 2021; Sun et al., 2012)
	Methanol truck	334.53	1.8000	0.1000	g/km	(Ingham, 2017; Xu and Peng, 2016)

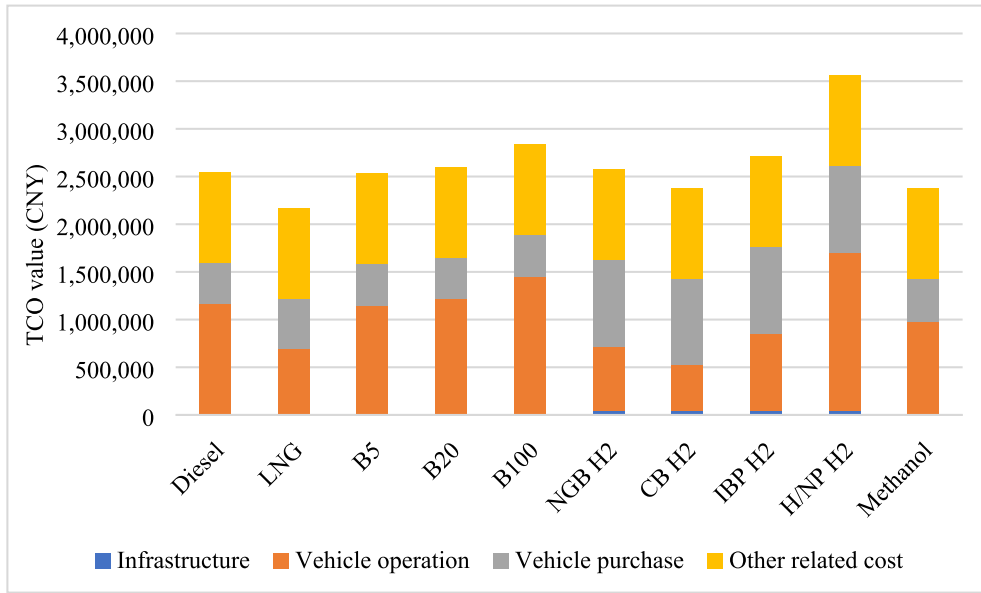


Fig. 3. TCO comparison of different alternative fuel HDTs.

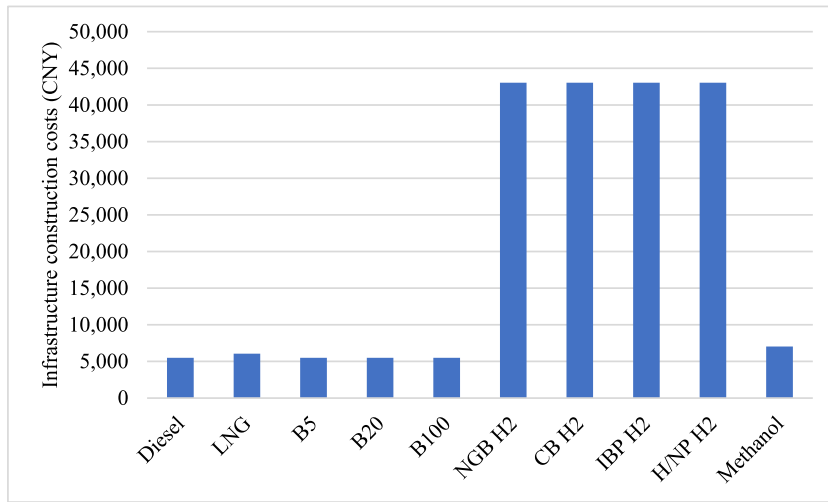


Fig. 4. Comparison of refueling infrastructure costs for different alternative fuel HDTs.

The GHG emissions of WTW of different HDTs are shown in Fig. 5, the ranking from low to high is as follows: B20, B100, B5, LNG, H/NP H₂, IBP H₂, diesel, NGB H₂, methanol, and CB H₂. It is found that all biodiesel trucks have lower WTW GHG emissions than traditional diesel HDTs. The lower GHG emissions of biodiesel trucks are due to the emission reductions of 52.7%, 54.3%, and 49.6% for B5, B20, and B100 trucks compared to diesel trucks during the PTW stage. For H₂ICE trucks, the GHG emissions from hydrogen production determine their total WTW emissions. Therefore, IBP H₂ and H/NP H₂ HDTs from low-carbon energy have lower GHG emissions than traditional diesel trucks, while H₂ICE trucks with other hydrogen sources (NGB H₂ and CB H₂) have higher GHG emissions. Due to the high emission value in the WTP stage, the WTW GHG of methanol HDTs is higher than that of traditional diesel.

Fig. 6 illustrates the NO_x emissions of alternative fuel HDTs, and the ranking from low to high is as follows: H/NP H₂, CB H₂, IBP H₂, NGB H₂, methanol, LNG, B100, B20, diesel, and B5. It is found that H₂ICE trucks are at the lowest emission level in the WTW stage, which reflects the characteristics of near-zero emissions. Methanol HDTs have higher NO_x emissions in the WTP stage. Due to the clean combustion of methanol fuel, it has better NO_x emission reduction performance in the WTW stage, which is 42.7% lower than that of traditional diesel, while the NO_x emission of LNG HDTs are 24.3% lower than traditional diesel HDTs.

The WTW PM_{2.5} emissions of different HDTs are shown in Fig. 7, and the ranking from low to high is as follows: H/P H₂, NGB H₂, B100, LNG, B20, methanol, IBP H₂, CB H₂, diesel, and B5. Among H₂ICE trucks, H/NP H₂ and NGB H₂ trucks have better emission reduction performance of PM_{2.5}. Methanol trucks have higher WTW PM_{2.5} emissions because coal is the main feedstock for methanol

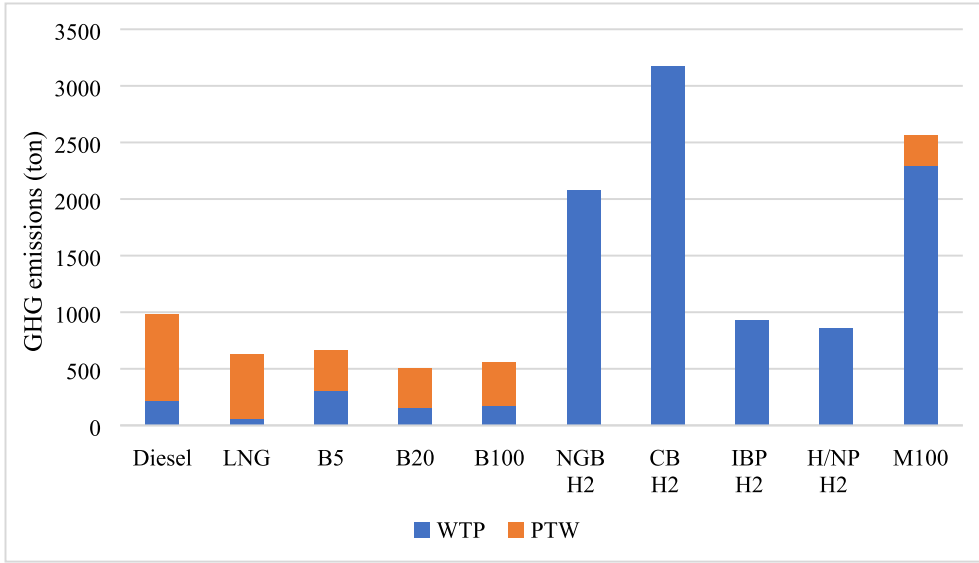


Fig. 5. Comparison of different alternative fuel HDTs' GHG emissions in the WTW cycle.

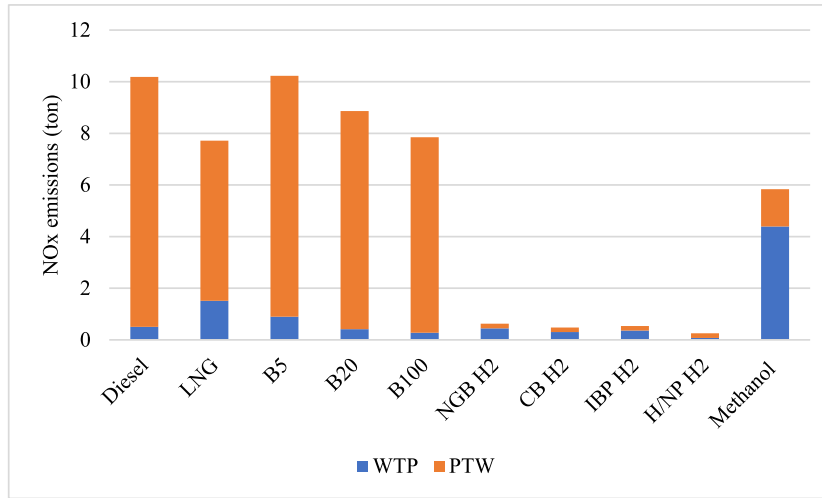


Fig. 6. Comparison of WTW NOx emissions for different alternative fuel HDTs.

production in China, and many particulate matter emissions are generated during the treatment process. Methanol HDTs have a low PM_{2.5} emission value in the PTW stage, which is only 24.4% of the traditional diesel HDTs. B20 and B100 trucks have brilliant WTW PM_{2.5} emissions performance, which is reduced by 48.6% and 76.2% respectively compared with traditional diesel. However, the PM_{2.5} emissions value of B5 HDTs is higher than that of traditional diesel. LNG HDTs have the third-best performance due to their low operating emissions, which are only higher than H/NP H₂ and NGB H₂.

4.3. Carbon-abatement benefit of different alternative fuel HDTs

Based on the TCO and WTW emission results of traditional diesel and alternative fuel HDTs (Section 4.1 and Section 4.2), the CAB results are calculated using Eq. (5). Higher CAB value correspond to a greater benefit obtained from operating alternative fuel vehicles. To some extent, a negative CAB value indicates that the alternative fuel has no comprehensive advantage over traditional diesel. As shown in Fig. 8, the ranking of benefits from low to high is as follows: H/NP H₂, B100, IBP H₂, NGB H₂, B20, CB H₂, B5, methanol, and LNG.

LNG HDT has the lowest TCO value (Fig. 3), and its GHG emission is lower than that of traditional diesel HDT (Fig. 5), so it has the highest CAB value. B5, CB H₂, and methanol HDTs have better benefits, which are 0.06, 0.002, and 0.07 CNY/km respectively. In the context of current vehicle costs and GHG emissions, using LNG and B5 HDTs to replace traditional diesel HDTs can best balance

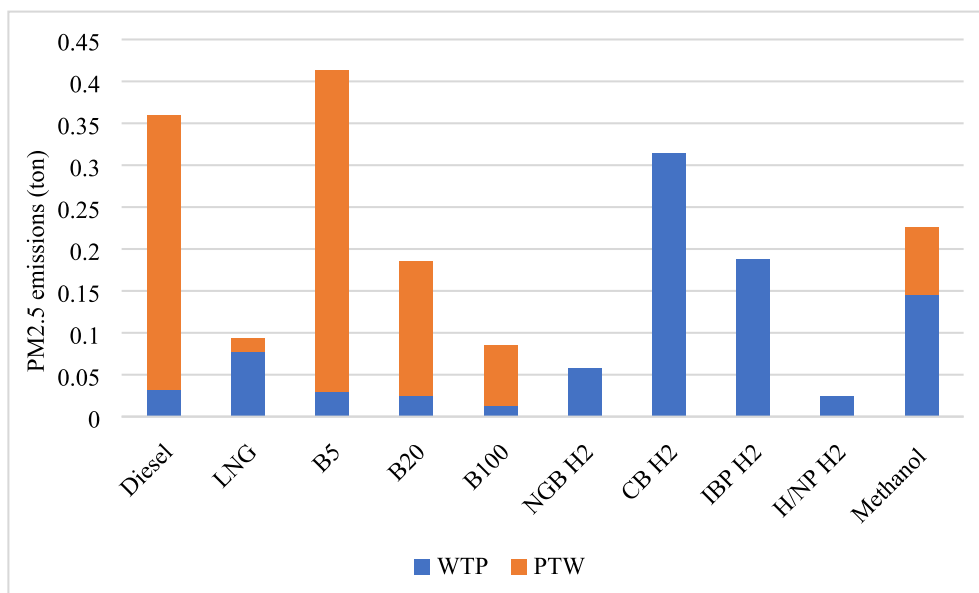


Fig. 7. Comparison of WTW PM2.5 emissions for different alternative fuel HDTs.

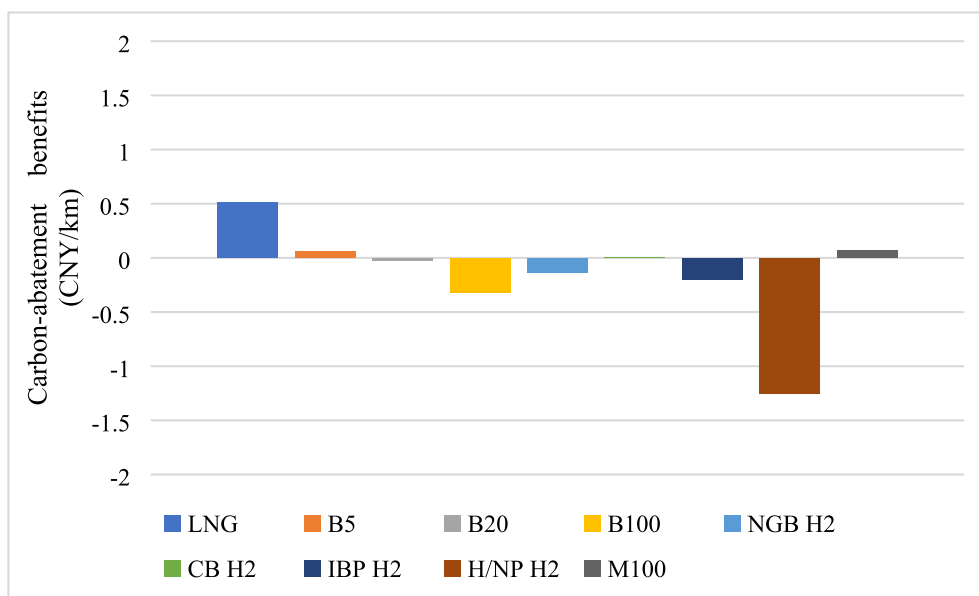


Fig. 8. CAB comparison of different alternative fuel HDTs.

economic and environmental performance. Except for CB H₂ trucks, the CAB of other H₂ICE trucks is negative, ranging from −1.25 to −0.14 CNY/km. This is because in terms of replacement costs, the TCO values of NGB H₂, IBP H₂, and H/NP H₂ HDT are higher than those of traditional diesel HDT (Fig. 3). It is worth noting that H/NP H₂ HDTs have the lowest CAB value among the four different fuel preparation pathways because of the immaturity of the technology associated with the electrolysis of water for hydrogen production and the expensive cost of industrial equipment. CB H₂ HDTs have the highest CAB value in the H₂ICE trucks because China has rich coal resources and the cost of hydrogen fuel produced by coal gasification is cheaper.

5. Scenario analysis

5.1. Scenario settings

Considering the uncertainty of alternative fuel development and future emission control plans, the scenario analysis of costs and emissions for each potential alternative fuel pathway is conducted using a staged prediction approach and existing data, which is divided into five stages: 2020, 2025, 2030, 2035 and 2040, to better guide the selection of the most promising alternative fuel.

As the increase in land and labor costs caused by the growth of the Chinese economy, the construction of traditional diesel fueling stations cannot receive financial subsidies from the government. Further improvement of the thermal efficiency of diesel engines is very difficult because the development has reached a bottleneck. The study assumes the cost of traditional diesel HDTs will increase by 2% at each stage. Based on the Paris Agreement and China's goal of achieving carbon peak neutrality, the study sets the emission of traditional diesel HDTs to be reduced by 2% at each stage.

In 2020, China extracted 192.5 billion cubic meters of natural gas, a year-on-year increase of 9.8%, and continued to promote *the seven-year action plan for increasing reserves and production* (National Energy Administration, China, 2021). Meanwhile, LNG, as a widely used alternative fuel in China's transport sector, has relatively mature infrastructure construction. Based on the development of the natural gas industry and the advancement of LNG vehicle technology, the study assumes a 5% reduction in the cost and emissions of LNG HDTs at each stage.

The Chinese government has formulated relevant policies to develop the biodiesel industry, exempting biodiesel products that meet the national standards from consumption tax and a 70% value-added tax rebate (Li et al., 2021). The study sets the cost of biodiesel HDTs to be reduced by 3% per stage. The further development of biodiesel preparation processes and biodiesel internal combustion engines will reduce their associated emissions. The study assumes a 4% reduction in emissions from biodiesel HDTs at each stage.

The Chinese government has proposed the basic principles for developing the hydrogen energy industry in relevant plans, including actively promoting technological innovation, focusing on breaking through hydrogen technology bottlenecks, and developing renewable energy hydrogen production (National Energy Administration, China, 2022). Under the background of the rapid development in the hydrogen energy industry, the industrial chain and hydrogen fuel production process will gradually mature. The study sets a cost and emission reduction of 8% per stage for hydrogen fuel heavy trucks.

As the largest producer and marketer of methanol in the world, China's methanol industry produced 49.84 million tons in 2020, approximately 49% of global methanol production (CPCIF, 2021). With the support of the mature methanol industry, the Chinese government has promoted the application of methanol vehicles in some regions and paid attention to the coordinated development of methanol fuel production, transmission, distribution, and refueling. Based on the development trend of methanol vehicles, the study sets the cost and emission of methanol HDTs to be reduced by 2% and 5% respectively.

5.2. Result and analysis

Combined with the above relevant parameter settings, the CAB results of different alternative fuel HDTs under scenario analysis are shown in Table 3. LNG HDTs have the highest CAB throughout the analysis cycle, based on their significant cost advantages compared to traditional diesel HDTs. By 2025, CB H₂ HDTs will surpass methanol HDTs, reaching the second-highest CAB value. By 2030 and 2035, all alternative fuel HDTs except for H/NP H₂ trucks will have positive CAB values, which means they will have better environment-economic benefits compared to the cost of replacing traditional diesel HDTs. The CAB results for 2040 are all positive, with the following rankings from lowest to highest: H/NP H₂, methanol, B100, B20, B5, IBP H₂, NGB H₂, CB H₂, and LNG. Among them, H/NP H₂ HDT has the lowest CAB among the four hydrogen fuel paths because of the high hydrogen fuel preparation cost, while CB H₂ HDTs prepared by coal gasification have lower vehicle operating costs and therefore have a higher CAB value. Compared with the CAB in 2020, the CAB value of hydrogen fuel HDTs has increased the most in other stages, followed by biodiesel, methanol, and LNG HDTs. In the future, more mature emission reduction technologies and lower full lifecycle vehicle costs are needed to achieve higher CAB values.

Through environment-economic and scenario analysis of potential alternative fuel HDTs, this study provides suggestions for the low-carbon development of heavy-duty freight in cold regions of China. At present, LNG HDTs can be used to replace traditional diesel HDTs to effectively control NO_x and PM_{2.5} emissions. Throughout the entire scenario analysis cycle, biodiesel trucks, especially B5

Table 3
CAB values of different alternative fuel HDTs from 2020 to 2040 (Unit: CNY/km).

Alternative fuel HDTs	2020	2025	2030	2035	2040
LNG	0.51	0.71	0.90	1.09	1.28
B5	0.06	0.21	0.50	0.76	1.02
B20	-0.03	0.14	0.42	0.69	0.95
B100	-0.32	-0.15	0.16	0.45	0.72
NGB H ₂	-0.14	0.20	0.51	0.81	1.09
CB H ₂	0.00	0.33	0.63	0.92	1.19
IBP H ₂	-0.20	0.14	0.46	0.76	1.04
H/NP H ₂	-1.25	-0.83	-0.43	-0.06	0.29
Methanol	0.07	0.20	0.34	0.47	0.60

HDTs, have shown good carbon reduction effects. It is worth noting that although biodiesel can improve combustion efficiency and reduce emissions, its calorific value is lower than diesel, which may lead to an increase in carbon dioxide emissions. The impact of the biodiesel blending ratio on emissions is complex and requires special attention. From 2020 to 2040, methanol trucks have a positive CAB value and can effectively replace traditional diesel HDT in mature regions of the methanol industry. By 2040, all alternative fuel HDTs will have better environment-economic benefits based on a cost and emission reduction of 8% per stage for hydrogen fuel HDTs. More economical hydrogen production methods in the future will bring about the rapid development of H₂ HDTs.

6. Conclusion

Based on the Paris Agreement and China's carbon abatement targets, the environment-economic analysis of alternative fuel internal combustion engine HDTs was conducted based on the comparison of CAB, TCO, and WTW emissions in the study. The alternative fuels selected include LNG, biodiesel (B5, B20, B100), hydrogen (NGB H₂, CB H₂, IBP H₂, H/NP H₂) and methanol. The relevant findings obtained from this study are as follows:

- LNG HDTs have the best economic performance due to the lower vehicle operating cost, followed by methanol. The economic performance of biodiesel depends on the mixture proportions of biofuel, and the TCO of B5 HDTs is lower than that of B20 and B100. Hydrogen is affected by the production pathways, and CB H₂ has a significant cost advantage compared to other hydrogen production methods.
- On the environmental side, the PTW GHG emission reductions of B5, B20, and B100 HDTs are 52.7%, 54.3%, and 49.6%, reflecting their carbon-abatement capabilities during vehicle operation. H/NP H₂ HDTs perform better in terms of WTW GHG emissions in the four hydrogen preparation pathways. H₂ICE trucks have the lowest environmental impact on NO_x emissions, while only B5 HDTs' PM_{2.5} emissions are higher than traditional diesel trucks.
- CAB indicators of different alternative fuel HDTs were calculated and found that LNG, B5, and methanol HDTs all have positive CAB values, which means that these HDTs can bring net environment-economic benefits by replacing diesel HDTs at the current condition of technology and regulation. Except for CB H₂, the CAB values of H₂ICE trucks on other hydrogen preparation paths are all negative.

In addition, a predictive scenario analysis regarding CAB values was performed. LNG HDTs can be used to replace traditional diesel HDTs at the current stage to effectively control NO_x and PM_{2.5} emissions. Throughout the scenario analysis cycle, biodiesel trucks have a good carbon reduction effect. The methanol trucks have a positive CAB value and can effectively replace traditional diesel HDTs in regions where the methanol industry is mature.

As the development and application of alternative fuel HDTs is a dynamic process, one of the challenges of this study is the potential uncertainty of the results caused by using representative data from outside China. Another challenge is the fluctuating value of emissions from upstream fuel production. Moreover, limited by the difficulty in quantifying the economic benefits of NO_x and PM_{2.5} emissions abatement, the study only considered the CAB value of alternative fuel HDTs. Fortunately, the framework proposed in this study can be applied to future research by adjusting system boundaries and data. For example, expanding the framework by considering the benefits of NO_x and PM_{2.5} emissions abatement. Considering China's carbon peaking and neutrality goals, other issues in the transport sector should be explored in future research, such as the driving performance of alternative fuel HDTs in different scenarios, the impact of purchase subsidies, the fluctuation of fuel prices, and carbon abatement policies. Future research endeavors could explore a wider range of clean trucks, especially the choice between electric trucks and alternative fuel trucks.

CRediT authorship contribution statement

Xizhao Zhang: Writing – original draft, Formal analysis, Conceptualization. **Zhenhong Lin:** Writing – review & editing, Validation. **Han Hao:** Writing – review & editing, Validation. **Xu Hao:** Writing – review & editing, Validation. **Zhichao Wang:** Writing – review & editing, Validation. **Shunxi Li:** Validation, Supervision, Project administration, Funding acquisition.

Data availability

Data will be made available on request.

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